

ALY 6010 Prob Theory and Intro Stat – Report

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Module4 - R Practice

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INTRODUCTION

The aim of this analysis is to perform two-sample t-test for two separate dataset “**cats**” and “**meditation workshop**”.

Scenario 1: Do male and female cat samples have the same bodyweight ("Bwt")?

For this scenario, we will conduct a two-sample t-test with unequal variance to compare the bodyweights of male and female cats using the "cats" dataset. This will involve separating the dataset into male and female samples and then performing the appropriate statistical test.

Scenario 2: Does meditation improve sleeping quality?

For this scenario, we will utilize a paired t-test to assess whether meditation has a significant effect on sleeping quality. This will involve comparing the sleeping quality scores of the same individuals before and after the meditation workshop.

Installing required libraries

```
library(MASS)
```

PART-1

Creating separate vectors for male and female cat bodyweights

Creating separate vectors for male and female cat bodyweights is essential for conducting statistical analyses to determine if there are significant differences in bodyweights between the two groups

```
male <- subset(cats, subset = (cats$Sex == "M"))$Bwt
male

## [1] 2.0 2.0 2.1 2.2 2.2 2.2 2.2 2.2 2.2 2.2 2.2 2.2 2.3 2.4 2.4 2.4 2.4 2.4 2
.5 2.5
## [20] 2.5 2.5 2.5 2.5 2.5 2.5 2.6 2.6 2.6 2.6 2.6 2.6 2.6 2.7 2.7 2.7 2.7 2.7 2
.7 2.7
## [39] 2.7 2.7 2.8 2.8 2.8 2.8 2.8 2.8 2.8 2.8 2.9 2.9 2.9 2.9 2.9 3.0 3.0 3.0 3
.0 3.0
## [58] 3.0 3.0 3.0 3.0 3.1 3.1 3.1 3.1 3.1 3.1 3.1 3.1 3.2 3.2 3.2 3.2 3.2 3.2 3
.3 3.3
## [77] 3.3 3.3 3.4 3.4 3.4 3.4 3.4 3.4 3.5 3.5 3.5 3.5 3.5 3.5 3.6 3.6 3.6 3.6 3
.7 3
.8 3.8
## [96] 3.9 3.9

female <- subset(cats, subset = (cats$Sex == "F"))$Bwt
female

## [1] 2.0 2.0 2.0 2.1 2.1 2.1 2.1 2.1 2.1 2.1 2.1 2.1 2.1 2.2 2.2 2.2 2.2 2.2 2
.2 2.3
## [20] 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.3 2.4 2.4 2.4 2.4 2.5 2
.5 2
.6 2.6
## [39] 2.6 2.7 2.7 2.7 2.7 2.9 2.9 2.9 3.0 3.0
```

Constructing two sample t-test for unequal variance

(a) Determining sample size

```
sample.size.1 <- length( male )
cat( "Sample size of male:",
     round( sample.size.1, 5 ) )

## Sample size of male: 97

sample.size.2 <- length( female )
cat( "Sample size of female:",
     round( sample.size.2, 5 ) )

## Sample size of female: 47
```

(b) Determining degree of freedom

```
degrees.of.freedom.1 <- sample.size.1 - 1
degrees.of.freedom.1

## [1] 96

degrees.of.freedom.2 <- sample.size.2 - 1
degrees.of.freedom.2

## [1] 46

total.degrees.of.freedom <-
  degrees.of.freedom.1 + degrees.of.freedom.2
cat("Degree of freedom ",round(total.degrees.of.freedom,5))

## Degree of freedom 142
```

(c) Significance level

```
significance_level <- 0.05
significance_level

## [1] 0.05
```

(d) lower critical value

```
lower.critical.value <-
  qt( significance_level / 2, total.degrees.of.freedom)
cat( "Lower critical value:",
  round( lower.critical.value, 5 ) )

## Lower critical value: -1.97681
```

(e) upper critical value

```
upper.critical.value <-
  qt( 1 - significance_level / 2, total.degrees.of.freedom)

cat( "Upper critical value:",
  round( upper.critical.value, 5 ) )

## Upper critical value: 1.97681
```

(f) Graphing density curve

```
# defining lower and upper critical values

# Plot the t-density curve
curve(
  dt(x, df = total.degrees.of.freedom),
  xlim = c(-10, 10),
  ylim = c(0, 0.5),
  main = "Two-sample hypothesis test",
  xlab = "t",
  ylab = "Density",
  lty = "solid",
  lwd = 3,
  col = "black"
)

# Shade the area to the left of the lower critical value
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -10, by = -0.01), -10),
  y = c(0, dt(seq(lower.critical.value, -10, by = -0.01), df = total.degrees.of.freedom), 0),
  col = "lightcoral"
)

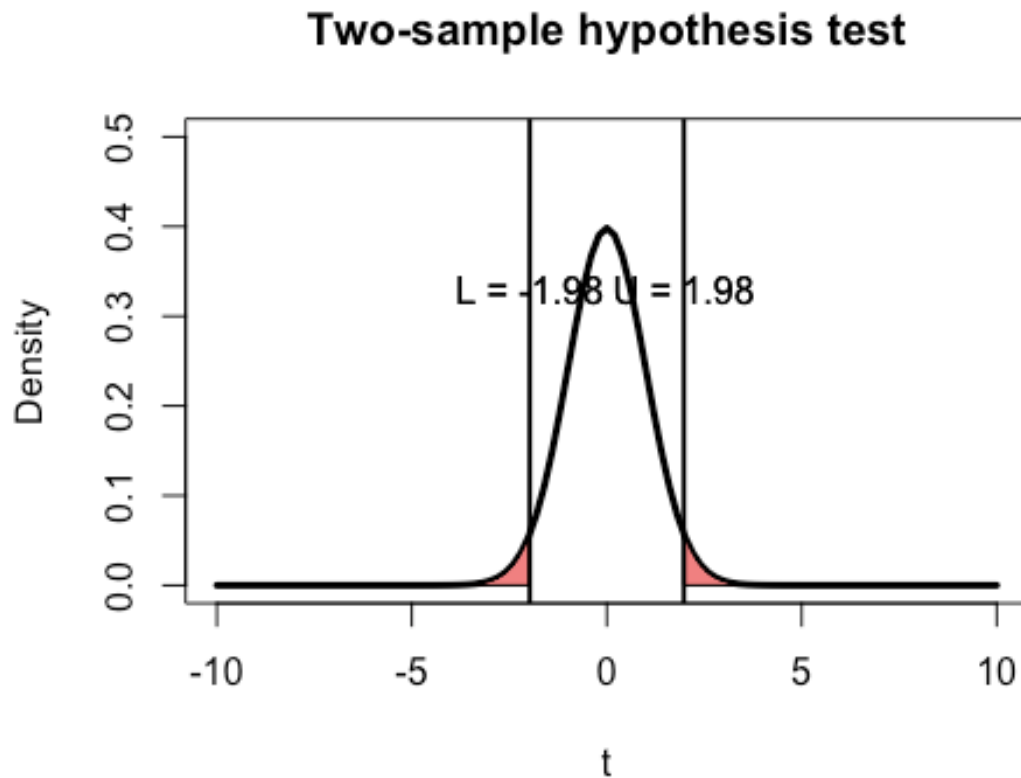
# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 10, by = 0.01), 10),
  y = c(0, dt(seq(upper.critical.value, 10, by = 0.01), df = total.degrees.of.freedom), 0),
  col = "lightcoral"
)

# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)

# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 3)), paste("U =", format(upper.critical.value, digits = 3)))
)

# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 3)), paste("U",
```

```
=", format(upper.critical.value, digits = 3)))  
)
```



Description: A density curve was created to display critical values at -1.98 and 1.98 , representing the lower and upper bounds of a significance level set at 0.05 . These critical values are indicated with black lines on the graph.

CONDUCTING TWO_SAMPLE T-TEST

(a) Calculating sample mean for male and female

```
sample.mean.1 <- mean( male )
cat( "Sample mean male:",
      round( sample.mean.1, 5 ) )

## Sample mean male: 2.9

sample.mean.2 <- mean( female )
cat( "Sample mean female:",
      round( sample.mean.2, 5 ) )

## Sample mean female: 2.35957
```

(b) Calculating sample variance for male and female

```
sample.variance.1 <- var( male )
cat( "Sample variance 1:",
      round( sample.variance.1, 5 ) )

## Sample variance 1: 0.21854

sample.variance.2 <- var( female )
cat( "Sample variance 2:",
      round( sample.variance.2, 5 ) )

## Sample variance 2: 0.07507
```

(c) Calculating S_p , the pooled estimate of the common population variance

```
weight.1 <-
  degrees.of.freedom.1 / total.degrees.of.freedom
weight.2 <-
  degrees.of.freedom.2 / total.degrees.of.freedom
pooled.variance.estimate <-
  (weight.1 * sample.variance.1) +
  (weight.2 * sample.variance.2)
cat( "Pooled variance estimate:",
      round( pooled.variance.estimate, 5 ) )

## Pooled variance estimate: 0.17206
```

(d) Calculating the t statistic for this data

```
t.statistic <-
  (sample.mean.1 - sample.mean.2) /
  sqrt( pooled.variance.estimate *
        (1/sample.size.1 + 1/sample.size.2) )
cat( "t statistic:",
      round( t.statistic, 5 ) )

## t statistic: 7.33067
```

(e) density curve of the test statistic under the null hypothesis

```
lower.critical.value <- -1.97681
upper.critical.value <- 1.97681

# Plot the t-density curve
curve(
  dt(x, df = total.degrees.of.freedom),
  xlim = c(-10, 10),
  ylim = c(0, 0.5),
  main = "Two-sample hypothesis test",
  xlab = "t",
  ylab = "Density",
  lty = "solid",
  lwd = 3,
  col = "black"
)

# Shade the area to the left of the lower critical value
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -10, by = -0.01), -10),
  y = c(0, dt(seq(lower.critical.value, -10, by = -0.01), df = total.degrees.of.freedom), 0),
  col = "lightcoral"
)

# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 10, by = 0.01), 10),
  y = c(0, dt(seq(upper.critical.value, 10, by = 0.01), df = total.degrees.of.freedom), 0),
  col = "lightcoral"
)

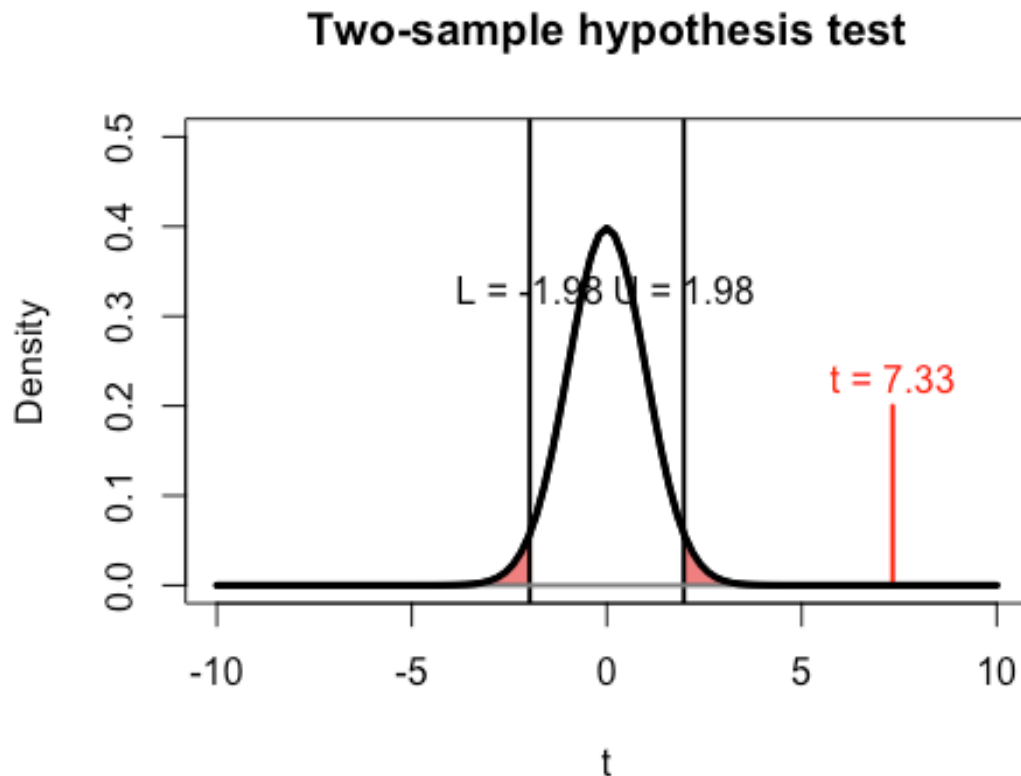
# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)

# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 3)), paste("U =", format(upper.critical.value, digits = 3)))
)
segments(t.statistic, 0, t.statistic, 0.2,
  lty = "solid", lwd = 2, col = "red")

text( t.statistic, 0.23,
  paste( "t =", round(t.statistic, 2) ), col = "red")
```

```
segments(-10, 0, 10, 0,
         lty = "solid", lwd = 2, col = "gray50")

curve(
  dt(x, total.degrees.of.freedom),
  lty = "solid", lwd = 3,
  col = "black", add = TRUE)
```



Description: A density curve was created to display critical values at -1.98 and 1.98 , representing the lower and upper bounds of a significance level set at 0.05 . These critical values are indicated with black lines on the graph. Additionally, the t-statistic value of 7.33 is plotted with red line. Notably, the t-value falls outside the bell curve, suggesting a significant deviation from the expected distribution under the null hypothesis.

(f) Confidence interval endpoint

```
lower.confidence.interval.endpoint <-  
  (sample.mean.1 - sample.mean.2) +  
  qt(0.025, total.degrees.of.freedom) *  
  sqrt( pooled.variance.estimate * (1/sample.size.1 + 1/sample.size.2) )  
  
cat( "Lower confidence interval estimate:",  
      round( lower.confidence.interval.endpoint, 5 ) )  
  
## Lower confidence interval estimate: 0.39469  
  
##upper  
upper.confidence.interval.endpoint <-  
  (sample.mean.1 - sample.mean.2) +  
  qt(0.975, total.degrees.of.freedom) *  
  sqrt( pooled.variance.estimate * (1/sample.size.1 + 1/sample.size.2) )  
  
cat( "Upper confidence interval estimate:",  
      round( upper.confidence.interval.endpoint, 5 ) )  
  
## Upper confidence interval estimate: 0.68616
```

(h) Performing Two sample t-test

Null Hypothesis (H0): The null hypothesis states that there is no difference in the mean bodyweight between male and female cats

H0: $\mu_{\text{male}} = \mu_{\text{female}}$

Where μ_{male} represents the bodyweight of male cats and μ_{female} represents the bodyweight of female cats.

Alternative Hypothesis (H1): The alternative hypothesis, which is what we are testing for, suggests that there is a difference in the mean bodyweight between male and female cats. It can be stated as:

H1: $\mu_{\text{male}} \neq \mu_{\text{female}}$

Collecting the data from "Cats" dataset, a two-sample t-test is conducted using the collected "Bodyweight" data and the specified null hypothesis expected value ($\mu_{\text{male}} = \mu_{\text{female}}$).

```
t.test(
  male,
  female,
  var.equal = FALSE
)

##
##  Welch Two Sample t-test
##
## data:  male and female
## t = 8.7095, df = 136.84, p-value = 8.831e-15
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.4177242 0.6631268
## sample estimates:
## mean of x mean of y
##  2.900000  2.359574
```

RESULT OF TWO SAMPLE T-TEST

Given the low p-value ($p \approx 8.831e-15$), which is much less than the significance level of 0.05, there is strong evidence to reject the null hypothesis. Therefore, we can conclude that there is a statistically significant difference in bodyweight between male and female cats.

(i) Calculating p-value

```
result <- t.test(male,female)
p_value <- result$p.value
p_value

## [1] 8.831034e-15
```

PART-2 [Meditation workshop]

1) (Significance level 0.05) :

Sleeping quality scores before the workshop

```
before <- c(4.6, 7.8, 9.1, 5.6, 6.9, 8.5, 5.3, 7.1, 3.2, 4.4)
before

## [1] 4.6 7.8 9.1 5.6 6.9 8.5 5.3 7.1 3.2 4.4
```

Sleeping quality scores after the workshop

```
after <- c(6.6, 7.7, 9.0, 6.2, 7.8, 8.3, 5.9, 6.5, 5.8, 4.9)
after

## [1] 6.6 7.7 9.0 6.2 7.8 8.3 5.9 6.5 5.8 4.9
```

Constructing two sample t-test

(a) determining sample size

```
sample.size.1 <- length( before )
cat( "Sample size of before:",
      round( sample.size.1, 5 ) )

## Sample size of before: 10

sample.size.2 <- length( after )
cat( "Sample size of after:",
      round( sample.size.2, 5 ) )

## Sample size of after: 10
```

(b) determining degree of freedom

```
degrees.of.freedom.1 <- sample.size.1 - 1
degrees.of.freedom.1

## [1] 9

degrees.of.freedom.2 <- sample.size.2 - 1
degrees.of.freedom.2

## [1] 9

total.degrees.of.freedom <-
  degrees.of.freedom.1 + degrees.of.freedom.2
cat("Degree of freedom ",round(total.degrees.of.freedom,5))

## Degree of freedom 18
```

(c) lower critical value

```
significance_level <- 0.05
significance_level

## [1] 0.05

lower.critical.value <-
  qt( significance_level / 2, total.degrees.of.freedom)
cat( "Lower critical value:",
      round( lower.critical.value, 5 ) )

## Lower critical value: -2.10092
```

(d) upper critical value

```
upper.critical.value <-
  qt( 1 - significance_level / 2, total.degrees.of.freedom)

cat( "Upper critical value:",
      round( upper.critical.value, 5 ) )

## Upper critical value: 2.10092
```

(e) Graphing density curve Plot the t-density curve Define lower and upper critical values

```
curve(
  dt(x, df = total.degrees.of.freedom),
  xlim = c(-4, 4),
  ylim = c(0, 0.5),
  main = "Two-sample hypothesis test",
  xlab = "t",
  ylab = "Density",
  lty = "solid",
  lwd = 3,
  col = "black"
)

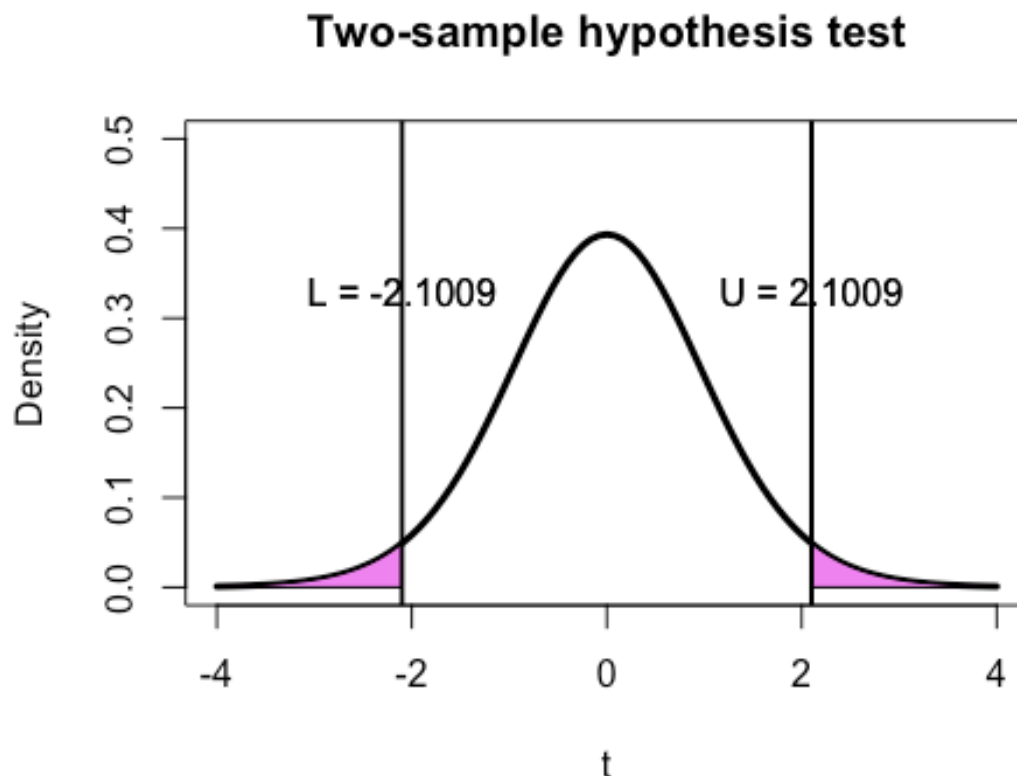
# Shade the area to the left of the lower critical value
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -4, by = -0.01), -4),
  y = c(0, dt(seq(lower.critical.value, -4, by = -0.01), df = total.degrees.o
f.freedom), 0),
  col = "violet"
)

# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 4, by = 0.01), 4),
  y = c(0, dt(seq(upper.critical.value, 4, by = 0.01), df = total.degrees.of.
freedom), 0),
  col = "violet"
```

```
)

# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)

# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)
# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)
```



Description: A density curve was created to display critical values at -2.1009 and 2.1009, representing the lower and upper bounds of a significance level set at 0.05. These critical values are indicated with black lines on the graph.

CONDUCTING TWO_SAMPLE T-TEST

(a) Calculating sample mean for male and female

```
sample.mean.1 <- mean( before )
cat( "Sample mean before:",
      round( sample.mean.1, 5 ) )

## Sample mean before: 6.25

#female
sample.mean.2 <- mean( after )
cat( "Sample mean after:",
      round( sample.mean.2, 5 ) )

## Sample mean after: 6.87
```

(b) Calculating sample variance for male and female

```
sample.variance.1 <- var( before )
cat( "Sample variance 1:",
      round( sample.variance.1, 5 ) )

## Sample variance 1: 3.72278

sample.variance.2 <- var( after )
cat( "Sample variance 2:",
      round( sample.variance.2, 5 ) )

## Sample variance 2: 1.64011
```

(c) Calculating S_p , the pooled estimate of the common population variance

```
weight.1 <-
  degrees.of.freedom.1 / total.degrees.of.freedom
weight.2 <-
  degrees.of.freedom.2 / total.degrees.of.freedom
pooled.variance.estimate <-
  (weight.1 * sample.variance.1) +
  (weight.2 * sample.variance.2)
cat( "Pooled variance estimate:",
      round( pooled.variance.estimate, 5 ) )

## Pooled variance estimate: 2.68144
```

(d) Calculating the t statistic for this data

```
t.statistic <-
  (sample.mean.1 - sample.mean.2) /
  sqrt( pooled.variance.estimate *
        (1/sample.size.1 + 1/sample.size.2) )
cat( "t statistic:",
      round( t.statistic, 5 ) )
```

```
## t statistic: -0.84663
```

(e) Density curve of the test statistic under the null hypothesis

```
lower.critical.value <- -2.10092 upper.critical.value <- 2.10092
```

```
curve(
  dt(x, df = total.degrees.of.freedom),
  xlim = c(-4, 4),
  ylim = c(0, 0.5),
  main = "Two-sample hypothesis test",
  xlab = "t",
  ylab = "Density",
  lty = "solid",
  lwd = 3,
  col = "black"
)

# Shade the area to the left of the lower critical value
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -4, by = -0.01), -4),
  y = c(0, dt(seq(lower.critical.value, -4, by = -0.01), df = total.degrees.o
f.freedom), 0),
  col = "violet"
)

# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 4, by = 0.01), 4),
  y = c(0, dt(seq(upper.critical.value, 4, by = 0.01), df = total.degrees.of.
freedom), 0),
  col = "violet"
)

# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)

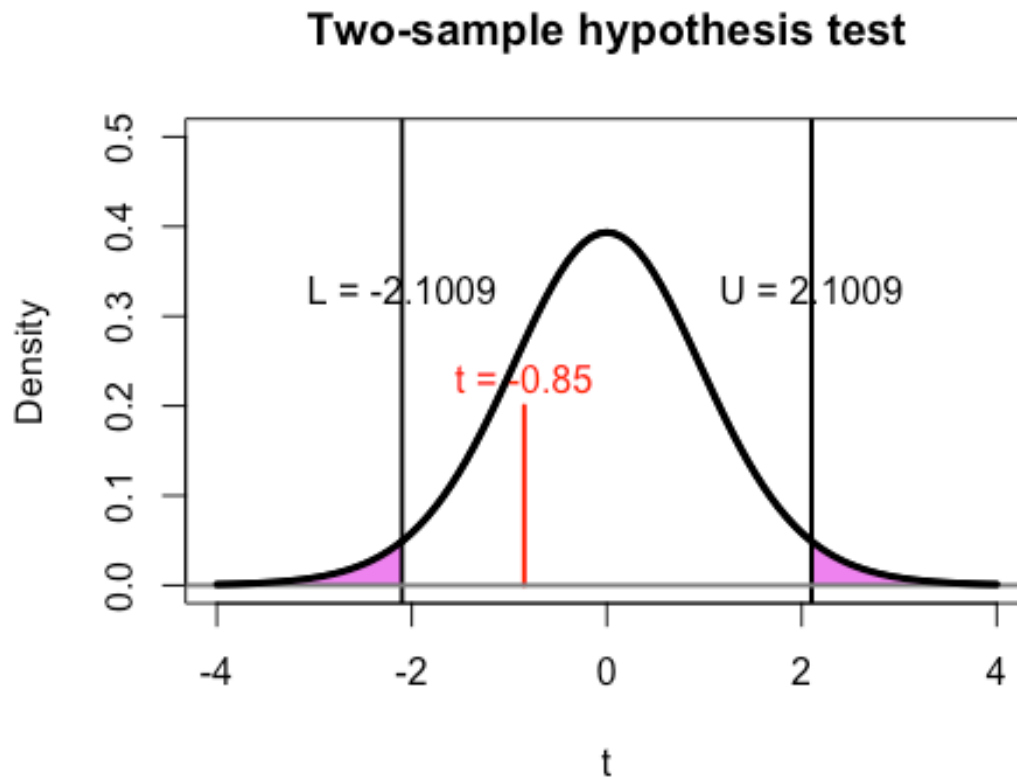
# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)
segments(t.statistic, 0, t.statistic, 0.2,
  lty = "solid", lwd = 2, col = "red")

text( t.statistic, 0.23,
  paste( "t =", round(t.statistic, 2) ), col = "red")

segments(-10, 0, 10, 0,
```

```
lty = "solid", lwd = 2, col = "gray50")

curve(
  dt(x, total.degrees.of.freedom),
  lty = "solid", lwd = 3,
  col = "black", add = TRUE)
```



Description: A density curve was created to display critical values at -2.1009 and 2.1009, representing the lower and upper bounds of a significance level set at 0.05. These critical values are indicated with black lines on the graph. Additionally, the t-statistic value of -0.85 is plotted with red line. Notably, the t-value falls **INSIDE** the bell curve.

- (f) perform a two-sided test of the null hypothesis $h_0: \mu_1 = \mu_2$ using a significance level of $\alpha = 0.05$

```
lower.confidence.interval.endpoint <-
  (sample.mean.1 - sample.mean.2) +
  qt(0.025, total.degrees.of.freedom) *
  sqrt( pooled.variance.estimate * (1/sample.size.1 + 1/sample.size.2) )
```

```
cat( "Lower confidence interval estimate:",
      round( lower.confidence.interval.endpoint, 5 ) )

## Lower confidence interval estimate: -2.15854

##upper
upper.confidence.interval.endpoint <-
  (sample.mean.1 - sample.mean.2) +
  qt(0.975, total.degrees.of.freedom) *
  sqrt( pooled.variance.estimate * (1/sample.size.1 + 1/sample.size.2) )

cat( "Upper confidence interval estimate:",
      round( upper.confidence.interval.endpoint, 5 ) )

## Upper confidence interval estimate: 0.91854
```

(h) Two sample t-test

Null Hypothesis (H0): There is no difference in sleep quality before and after the meditation workshop.

Alternative Hypothesis (H1): There is a difference in sleep quality before and after the meditation workshop.

```
{r Two sample t-test for sleeping evaluation} t.test( before,
after, var.equal = FALSE )
```

```
Welch Two Sample t-test

data: before and after
t = -0.84663, df = 15.641, p-value = 0.41
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -2.1753439  0.9353439
sample estimates:
mean of x mean of y
 6.25      6.87
```

Result of two sample t-test

This t-value (-0.84663) tells us how much the sleep quality changed after the meditation workshop compared to before. A negative value means the sleep quality tended to be a bit lower after the workshop, but not by much.

This is a measure of how likely it is that the changes in sleep quality we observed could have happened just by chance. Here, the p-value is 0.41, which is quite high. It means there's a pretty good chance that any differences in sleep quality we saw were just random fluctuations and not due to the meditation workshop.

(i) p-value calculate

```
result <- t.test(before,after)
p_value <- result$p.value
p_value

## [1] 0.4099699
```

PART-2 [Meditation workshop]

2) (Significance level 0.1)

```
significance_level <- 0.1
significance_level

## [1] 0.1
```

lower Critical value

```
lower.critical.value <-
  qt( significance_level / 2, total.degrees.of.freedom)
cat( "Lower critical value:",
      round( lower.critical.value, 5 ) )

## Lower critical value: -1.73406
```

Upper critical value

```
upper.critical.value <-
  qt( 1 - significance_level / 2, total.degrees.of.freedom)
cat( "Upper critical value:",
      round( upper.critical.value, 5 ) )

## Upper critical value: 1.73406
```

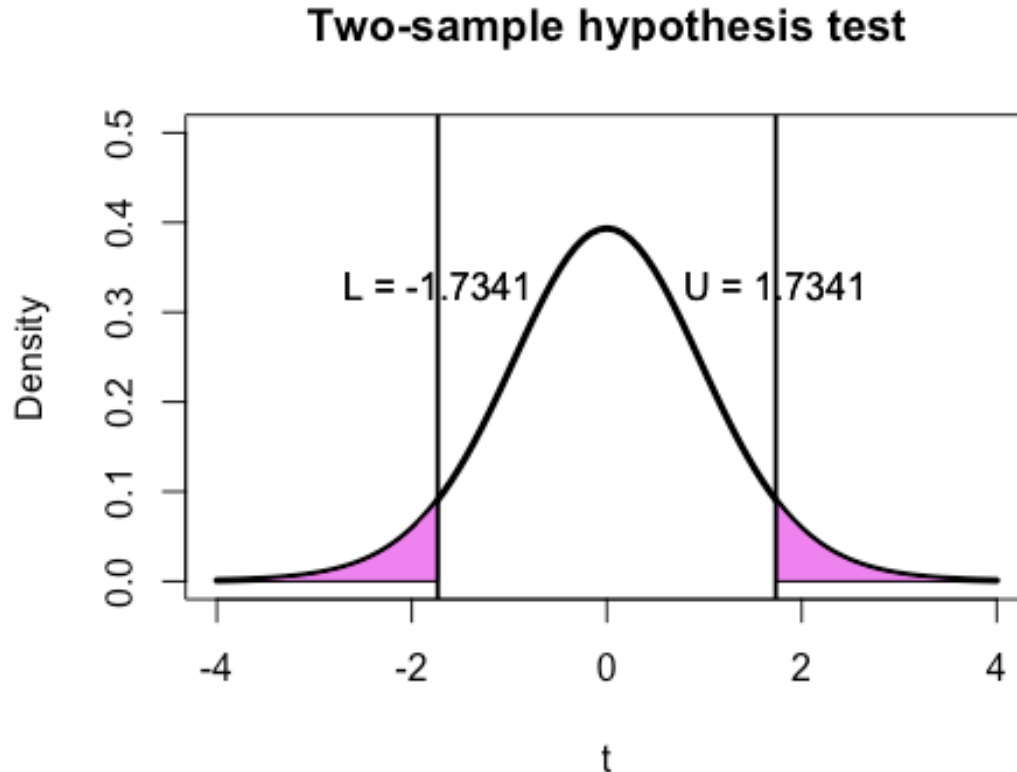
GRAPHING DENSITY CURVE

```
lower.critical.value <- -1.73406
upper.critical.value <- 1.73406
# Plot the t-density curve
curve(
  dt(x, df = total.degrees.of.freedom),
  xlim = c(-4, 4),
  ylim = c(0, 0.5),
  main = "Two-sample hypothesis test",
  xlab = "t",
  ylab = "Density",
  lty = "solid",
  lwd = 3,
  col = "black"
)
```

```

# Shade the area to the left of the lower critical value
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -4, by = -0.01), -4),
  y = c(0, dt(seq(lower.critical.value, -4, by = -0.01), df = total.degrees.o
f.freedom), 0),
  col = "violet"
)
# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 4, by = 0.01), 4),
  y = c(0, dt(seq(upper.critical.value, 4, by = 0.01), df = total.degrees.of.
freedom), 0),
  col = "violet"
)
# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)
# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)
# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)

```



Description: A density curve was created to display critical values at -1.7341 and 1.7341, representing the lower and upper bounds of a significance level set at 0.1. These critical values are indicated with black lines on the graph.

CONDUCTING TWO_SAMPLE T-TEST

Calculating the t statistic for this data

```
t.statistic <-
  (sample.mean.1 - sample.mean.2) /
  sqrt( pooled.variance.estimate *
        (1/sample.size.1 + 1/sample.size.2) )
cat( "t statistic:",
      round( t.statistic, 5 ) )

## t statistic: -0.84663
```

Density curve of the test statistic under the null hypothesis

```
lower.critical.value <- -1.73406
upper.critical.value <- 1.73406
curve(
  dt(x, df = total.degrees.of.freedom),
```

```

xlim = c(-4, 4),
ylim = c(0, 0.5),
main = "Two-sample hypothesis test",
xlab = "t",
ylab = "Density",
lty = "solid",
lwd = 3,
col = "black"
)
polygon(
  x = c(lower.critical.value, seq(lower.critical.value, -4, by = -0.01), -4),
  y = c(0, dt(seq(lower.critical.value, -4, by = -0.01), df = total.degrees.o
f.freedom), 0),
  col = "violet"
)
# Shade the area to the right of the upper critical value
polygon(
  x = c(upper.critical.value, seq(upper.critical.value, 4, by = 0.01), 4),
  y = c(0, dt(seq(upper.critical.value, 4, by = 0.01), df = total.degrees.of.
freedom), 0),
  col = "violet"
)

# Add vertical lines for critical values
abline(v = c(lower.critical.value, upper.critical.value), lwd = 2)

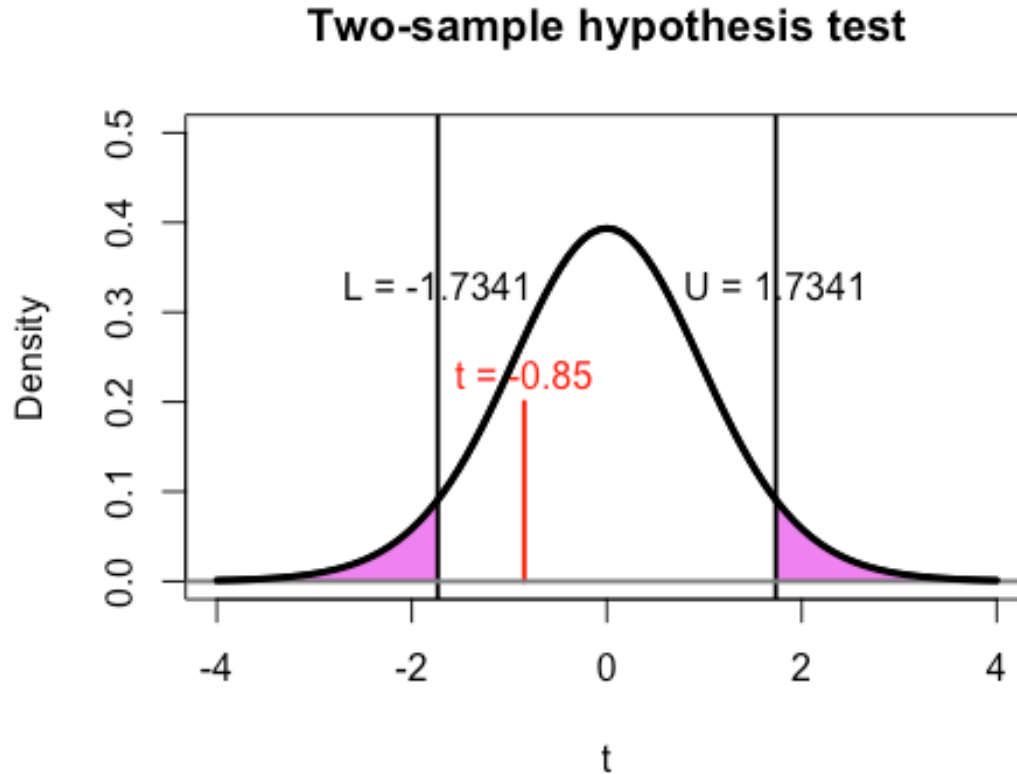
# Add text for critical values
text(
  x = c(lower.critical.value, upper.critical.value),
  y = 0.33,
  labels = c(paste("L =", format(lower.critical.value, digits = 5)), paste("U
=", format(upper.critical.value, digits = 5)))
)
segments(t.statistic, 0, t.statistic, 0.2,
  lty = "solid", lwd = 2, col = "red")

text( t.statistic, 0.23,
  paste( "t =", round(t.statistic, 2) ), col = "red")

segments(-10, 0, 10, 0,
  lty = "solid", lwd = 2, col = "gray50")

curve(
  dt(x, total.degrees.of.freedom),
  lty = "solid", lwd = 3,
  col = "black", add = TRUE)

```



Description: A density curve was created to display critical values at -1.7341 and 1.7341, representing the lower and upper bounds of a significance level set at 0.1. These critical values are indicated with black lines on the graph. Additionally, the t-statistic value of -0.85 is plotted with red line. Notably, the t-value falls **INSIDE** the bell curve.

Lower confidence interval endpoint (0.1 significance level)

```
lower.confidence.interval.endpoint <- (sample.mean.1 - sample.mean.2) + qt(0.
05, total.degrees.of.freedom) * sqrt(pooled.variance.estimate * (1/sample.siz
e.1 + 1/sample.size.2))
cat("Lower confidence interval estimate:", round(lower.confidence.interval.en
dpoint, 5), "\n")

## Lower confidence interval estimate: -1.88989
```

Upper confidence interval endpoint (0.1 significance level)

```
upper.confidence.interval.endpoint <- (sample.mean.1 - sample.mean.2) + qt(0.90, total.degrees.of.freedom) * sqrt(pooled.variance.estimate * (1/sample.size.1 + 1/sample.size.2))
cat("Upper confidence interval estimate:", round(upper.confidence.interval.endpoint, 5), "\n")

## Upper confidence interval estimate: 0.35427
```

Two sample t-test

Null Hypothesis (H0): There is no difference in sleep quality before and after the meditation workshop.

Alternative Hypothesis (H1): There is a difference in sleep quality before and after the meditation workshop.

```
t_test_result <- t.test(before, after, alternative = "two.sided", conf.level = 0.90)
t_test_result

##
## Welch Two Sample t-test
##
## data: before and after
## t = -0.84663, df = 15.641, p-value = 0.41
## alternative hypothesis: true difference in means is not equal to 0
## 90 percent confidence interval:
## -1.9003429 0.6603429
## sample estimates:
## mean of x mean of y
## 6.25 6.87
```

RESULT OF TWO SAMPLE WITH SIGNIFICANCE LEVEL 0.1

The data analysis shows that the average sleep quality before the intervention was 6.25, and after the intervention, it was 6.87. The t-value is -0.84663, with a p-value of 0.41, suggesting weak evidence against the null hypothesis. The 90% confidence interval for the difference in means ranges from -1.9003429 to 0.6603429, indicating that there isn't strong confidence that the intervention made a significant difference in sleep quality. Therefore, it's uncertain if the intervention, likely a meditation workshop, had a meaningful impact on improving sleep quality.

CONCLUSION

Based on the three sets of analyses conducted, the following conclusions can be drawn:

In the comparison of bodyweight between male and female cats, a low p-value ($p \approx 8.831e-15$), significantly lower than the significance level of 0.05, indicates strong evidence to reject the null hypothesis. This suggests a statistically significant difference in bodyweight between male and female cats.

However, regarding the impact of a meditation workshop on sleep quality, the obtained p-value is 0.41, which exceeds the typical significance level of 0.05. This suggests weak evidence against the null hypothesis, indicating that any observed differences in sleep quality could likely be due to random fluctuations rather than the workshop itself.

Additionally, the examination of sleep quality before and after the intervention reveals an average increase in sleep quality from 6.25 to 6.87. However, the associated t-value of -0.84663 and the p-value of 0.41 suggest weak evidence against the null hypothesis, failing to provide strong confidence that the intervention made a significant difference in sleep quality.